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PH4202: Advanced Statistical Mechanics

ASSIGNMENT: 5

Question 1: Homogenous Function

$$f(\lambda r) = g(\lambda) f(r) \quad \forall \lambda.$$

$$\begin{aligned} \text{Let } \lambda, \mu \in \mathbb{R} \text{ then } f(\lambda(\mu x)) &= g(\lambda) f(\mu x) \\ &= g(\lambda) f(x) g(\mu) \end{aligned}$$

But we also have

$$f((\lambda\mu)x) = g(\lambda\mu) f(x).$$

$$\text{Equating both we get: } g(\lambda\mu) = g(\lambda) g(\mu) \quad \forall \lambda, \mu$$

As $g(xy) = g(x) g(y) \quad \forall x, y$ let us choose $y_0 \in \mathbb{R}$

$$\Rightarrow \boxed{g(xy_0) = g(x) g(y_0)} \quad (*)$$

$$\begin{aligned} \text{Given } f \rightarrow \text{differentiable, choose } r=1 \Rightarrow f(\lambda) &= g(\lambda) f(1) \\ \Rightarrow g(x) &= f(x)/f(1) \end{aligned}$$

$\Rightarrow g$ is also differentiable

Differentiating both sides of (*) we have:

$$g'(xy_0) y_0 = g(y_0) g'(x)$$

let us put $x=1$: $g'(y_0) y_0 = g(y_0) g'(1)$

Since y_0 was arbitrary we can treat g to be a function of y_0 and integrating we get:

$$g(y_0) = A y_0^p \quad \text{where } g'(1) = p$$

Now, let us fix A . The expression for g gives us

$$g(1) = (g(1))^2 \Rightarrow g(1) = 0 \text{ or } g(1) = 1$$

$$\text{If } g(1) = 0 \Rightarrow A = 0 \Rightarrow g(x) = 0 \quad \forall x$$

$$\text{If } g(1) = 1 \Rightarrow A = 1 \Rightarrow g(x) = x^p$$

$$\therefore \boxed{g(x) = x^p}$$

$$\Rightarrow \boxed{f(\lambda r) = \lambda^p \cdot f(r)}$$

Differentiating w.r.t. r we get: $\lambda f'(\lambda r) = \lambda^p \cdot f'(r)$

$$\text{Take } r=1: f'(\lambda) = \lambda^{p-1} f'(1)$$

$$\text{Integrating w.r.t. } \lambda: f(\lambda) = \frac{f'(1) \lambda^p}{p} \rightarrow \boxed{f(r) = \left(\frac{f'(1)}{p} \right) r^p}$$

$$\therefore \boxed{a=p} \quad \boxed{c = \frac{f'(1)}{p} = f(1)}$$

Question 3:

m is a generalised homogeneous function

$$m(\lambda^{a_t} t, \lambda^{a_h} h) = \lambda m(t, h)$$

let us take $\lambda^{a_h} h = 1 \Rightarrow \lambda = |h|^{-1/a_h}$

$$\therefore |h|^{-1/a_h} m(t, h) = m(t h^{-a_t/a_h}, 1)$$

The R.H.S is a single function of $\left(\frac{t}{h^{a_t/a_h}}\right)$ so we can write

$$m(t, h) = |h|^{1/a_h} \mathcal{G}_{\pm}\left(\frac{t}{h^{a_t/a_h}}\right)$$

Defining $\theta = 1/a_h$ and $\frac{a_t}{a_h} = \Delta_2$ we get:

$$m(t, h) = |h|^{\theta} \mathcal{G}_{\pm}\left(\frac{t}{h^{\Delta_2}}\right)$$

let us take different limits. For $h \neq 0$ if we take $t=0$, i.e., $T=T_c$ then $m \sim h^{1/8}$

$$\Rightarrow m(0, h) = |h|^{\theta} \mathcal{G}_{\pm}(0) \Rightarrow \boxed{\theta = 1/8}$$

And now if we take the limit $\hbar \rightarrow 0$, $m \sim |t|^\beta$

But $\left(\frac{t}{\hbar^{\Delta_2}}\right) \rightarrow \infty$. We want m to be finite. Assume

$$G_{\pm}(x) \sim x^p \Rightarrow m(t, \hbar \rightarrow 0) = |\hbar|^\theta \times \frac{t^p}{\hbar^{\Delta_2 p}}$$

to cancel effect of $\hbar$ we have $\boxed{\theta = \Delta_2 p}$ and to follow the typical scaling, $\boxed{p = \beta}$ $\Rightarrow \Delta_2 = \theta/p = \frac{1}{\delta} \times \frac{1}{\beta}$

$$\therefore \boxed{\Delta_2 = \frac{1}{\beta \delta}} \rightarrow \boxed{\theta = \frac{1}{\beta \delta} \times \beta = \frac{1}{\delta}}$$

Thus we can write

$$m(t, \hbar) = |\hbar|^{1/\delta} \cdot G\left(\frac{t}{|\hbar|^{1/\beta \delta}}\right)$$